

Molecular Dynamics Simulations with GROMACS

Protein–Protein & Protein–Drug Interactions, and Simplified System Preparation with CHARMM-GUI

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What We'll Cover



SECTION 1

MD Simulation of Protein–Protein Interactions

The complete GROMACS pipeline: topology, box, solvation, ions, minimization, equilibration, and production.



SECTION 2

MD Simulation of Protein–Drug Interactions

Extending the workflow to a system where a small-molecule ligand is bound to the protein.



SECTION 3

CHARMM-GUI: Simplified System Preparation

Building the same simulation-ready system through a graphical, browser-based interface.

SECTION 1 OF 3

Molecular Dynamics Simulation of Protein–Protein Interactions

The complete GROMACS pipeline — from a raw structure file to a finished protein trajectory.





What Is Molecular Dynamics Simulation?

- **Newton's equations of motion** are solved for every atom, one small time step at a time.
$$F=ma$$
- **A force field** supplies the potential-energy (Potential energy depends on position of atom) function bonded and non-bonded terms used to compute the force on each atom.
- **A static structure becomes a trajectory:** a series of snapshots showing how the system moves.
- **Reveals what one structure cannot:** flexibility, conformational change, and the dynamics of binding.
- Here F is the net interaction force acting on an atom due to all the other atoms in the system.
- It is not an external force like pushing a box. Instead, it is the sum of all the molecular interactions described by the force field.

$\sim 2 \text{ fs} = 0.002 \text{ ps}$

Typical integration time step

$\text{ns} - \mu\text{s}$

Typical production simulation length



Introduction to GROMACS

- **GROMACS** — the GROningen MAchine for Chemical Simulations.
- **Free and open-source**, under continuous development since the early 1990s.
- **Purpose-built for biomolecules:** proteins, nucleic acids, lipids, and their complexes.
- **Exceptionally fast:** highly optimized for CPUs and GPUs, and scales cleanly across supercomputing clusters.
- **A short chain of commands** carries a structure file all the way to a finished trajectory — the rest of this section walks through each one.



Introduction to GROMACS

Table 1: The most important files for beginners used in GROMACS

File	Purpose	Easy way to remember
.gro	Coordinates of all atoms and box size	Where are the atoms?
.top	Complete description of the molecular system	What molecules are present?
.itp	Topology details for individual molecules or force field components	How is each molecule built?
.mdp	Molecular dynamics simulation settings	How should the simulation be run?



The GROMACS Simulation Workflow



Six stages, one pipeline — each stage prepares the system for the next.

Sections 2 and 3 build on exactly this sequence.



Generating a GROMACS-Readable Topology

- **Start from a structure file (.pdb)**, typically downloaded from the RCSB Protein Data Bank.
- **pdb2gmx** converts the coordinate file into GROMACS format and builds the topology.
- **The topology (.top)** defines every bond, angle, dihedral, atom type, and partial charge in the system.
- **Also produces** a GROMACS coordinate file (.gro) and a position-restraint file (.itp).
- **This step is also where** the force field is chosen — covered next.

❑ **Command: `gmx pdb2gmx -f AA.pdb -p AA.top -o AA.gro -ignh`**

1. Reads the **input protein structure** (AA.pdb).
2. Allows you to choose a **force field** and **water model**.
3. Ignores existing hydrogen atoms and adds new ones consistent with the selected force field.
4. Generates the **system topology** (AA.top).
5. Creates the **GROMACS coordinate file** (AA.gro), which is used in subsequent simulation steps.

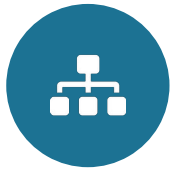


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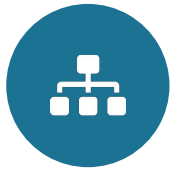
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Generating a GROMACS-Readable Topology

Table 2: Breakdown of GROMACS-Readable topology generating command

Command Part	Full Meaning	Function / Purpose
gmx	GROMACS executable	Starts the GROMACS program and provides access to all GROMACS tools.
pdb2gmx	PDB to GROMACS	Converts a PDB structure into GROMACS input files by assigning a force field, adding hydrogens, and generating topology and coordinate files.
-f AA.pdb	File (Input)	Specifies the input PDB file (AA . pdb) that GROMACS will read.
-p AA.top	Topology	Specifies the name of the topology file (AA . top) that GROMACS will generate. This file contains the molecular topology, force field information, and system composition.
-o AA.gro	Output	Specifies the name of the output coordinate file (AA . gro) that GROMACS will create in its native . gro format.
-ignh	Ignore Hydrogens	Ignores any hydrogens already present in the input PDB file and regenerates them according to the selected force field, ensuring correct atom names and protonation states.



Generating a GROMACS-Readable Topology

Table 3: Common Water models used in GROMACS

Water Model	Full Form	No. of Interaction Sites	Structure	Key Feature	Common Force Fields
SPC	Simple Point Charge	3	O + 2 H	Simple and computationally efficient	GROMOS
TIP3P	Transferable Intermolecular Potential – 3 Points	3	O + 2 H	Most widely used; compatible with CHARMM36 and AMBER	CHARMM36, AMBER
TIP4P	Transferable Intermolecular Potential – 4 Points	4	O + 2 H + 1 Dummy (M)	Uses an additional dummy charge to better represent the electrostatic properties of water	AMBER, OPLS, some CHARMM applications



Choosing a Force Field

A **force field** is the set of parameters and equations used to compute potential energy — bond lengths, angles, dihedrals, and non-bonded interactions.

AMBER

Proteins & nucleic acids; widely validated

CHARMM

Broad macromolecular coverage; pairs with CGenFF

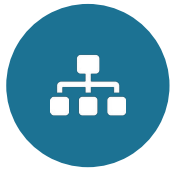
GROMOS

United-atom; compact and efficient

OPLS-AA

All-atom; strong small-molecule accuracy

- **Selection depends on** molecule type, water-model compatibility, and precedent in the literature.
- **Keep it consistent:** use the same force field family for the protein and any ligand (Section 2).



Generating a GROMACS-Readable Topology

Table 4: Comparison of All-Atom and United-Atom Force Fields

Feature	All-Atom (AA)	United-Atom (UA)
Hydrogen atoms	Every hydrogen atom is represented explicitly.	Hydrogens attached to carbon are merged with the carbon atom.
Number of atoms	More atoms	Fewer atoms
Computational cost	Higher	Lower
Accuracy	Higher	Slightly lower for some properties
Simulation speed	Slower	Faster
Force fields	CHARMM36, AMBER, OPLS-AA	GROMOS53A6, some GROMOS54A7/54A8 variants
Best suited for	Detailed biomolecular simulations	Large systems where computational efficiency is important



Defining the Simulation Box

- **editconf** places the solute inside a computational box.
- **Common shapes:** cubic, rhombic dodecahedron, or truncated octahedron.
- **Minimum image distance:** the molecule must sit far enough from the box edge (typically ≥ 1.0 – 1.2 nm) to avoid interacting with its own periodic copy.
- **A dodecahedral box** wraps the molecule more tightly than a cube — fewer water molecules, and a faster simulation.

❑ **Command: `gmx editconf -f AA.gro -o AA-newbox.gro -c -d 1.0 -bt cubic`**

1. Reads the protein coordinates from **AA.gro**.
2. Creates a **cubic simulation box**.
3. Centres the protein in the box.
4. Leaves at least **1.0 nm** of empty space between the protein and every side of the box.
5. Saves the new coordinates and box information in **AA-newbox.gro**.



Defining the Simulation Box

Table 5: Breakdown of the command defining Simulation Box

Command Part	Full Meaning	Function / Purpose
gmx	GROMACS executable	Starts the GROMACS program.
editconf	Edit Configuration	Modifies the coordinate file, such as changing the box size, shape, orientation, or centering the molecule.
-f AA.gro	File (Input)	Specifies the input coordinate file (AA.gro) containing the protein structure.
-o AA-newbox.gro	Output	Specifies the name of the output coordinate file with the newly defined simulation box.
-c	Center	Places the protein at the center of the simulation box.
-d 1.0	Distance	Sets a minimum distance of 1.0 nm (10 Å) between the protein and each face of the simulation box.
-bt cubic	Box Type	Creates a cubic simulation box around the protein.



Defining the Simulation Box

Table 6: Common types of Simulation Boxes used in MD Simulation while

Box Type	Option	Shape	Typical Use
Cubic	-bt cubic	Cube	Small or approximately spherical proteins; teaching examples
Dodecahedron	-bt dodecahedron	Truncated dodecahedron	Most common for globular proteins because it uses less solvent
Octahedron	-bt octahedron	Truncated octahedron	Compact alternative for roughly spherical molecules
Triclinic	-bt triclinic	General parallelepiped	Flexible box shape for specialized systems



Solvating the System

- **solvate** fills the remaining space in the box with explicit water molecules.
 - **Common water models:** SPC, SPC/E, and TIP3P.
 - **Recreates** the aqueous, physiological environment the biomolecule naturally occupies.
 - **Essential for realism:** proteins fold, move, and bind differently in vacuum than they do in water.
- ❑ **Command: `gmx solvate -cp AA-newbox.gro -cs tip4p.gro -o AA-solv.gro -p AA.top`**
1. Reads the protein and simulation box from **AA-newbox.gro**.
 2. Reads a pre-equilibrated box of water molecules from `spc216.gro`.
 3. Fills the empty space around the protein with water molecules.
 4. Removes any water molecules that overlap with the protein.
 5. Saves the solvated system as **AA-solv.gro**.
 6. Updates **AA.top** by adding the total number of water molecules.



Solvating the System

Table 7: Breakdown of the command solvating the system.

Command Part	Full Meaning	Function / Purpose
gmx	GROMACS executable	Starts the GROMACS program.
solvate	Solvate System	Adds water molecules to the simulation box surrounding the protein.
-cp AA-newbox.gro	Coordinate File (Protein)	Specifies the input coordinate file containing the protein already placed in the simulation box. (cp = coordinate protein/solute).
-cs spc216.gro	Coordinate Solvent	Specifies the solvent coordinate file (spc216 . gro) containing a pre-equilibrated box of TIP4P water molecules that will be used to fill the simulation box.
-o AA-solv.gro	Output	Specifies the name of the output coordinate file containing the protein surrounded by water molecules.
-p AA.top	Topology	Updates the topology file (AA . top) by adding the correct number of water molecules to the system.



Adding Ions to Neutralize the System

- **genion** replaces a small number of water molecules with ions — commonly Na⁺ and Cl⁻.
 - **Electroneutrality:** most biomolecules carry a net charge, and the periodic system as a whole must be neutral.
 - **Physiological concentration:** ions are usually added to roughly 0.15 M, not just to the minimum needed for neutrality.
 - **Prevents** artificial electrostatic forces that periodic boundary conditions would otherwise introduce.
- ❑ Command 1: `gmx grompp -f ions.mdp -c AA-solv.gro -p AA.top -o ions`
 - ❑ Command 2: `gmx genion -s ions.tpr -o AA-solv-ions.gro -p AA.top -pname NA -nname CL -neutral`

A .tpr (Portable Binary Run Input) file is the master input file for GROMACS. It combines the simulation parameters (.mdp), coordinates (.gro), topology (.top), and force field information into a single binary file that mdrun uses to perform the molecular dynamics simulation.



Adding Ions to Neutralize the System

❑ **Command 1**

1. Reads the simulation parameters from ions.mdp.
2. Reads the solvated protein coordinates from AA-solv.gro.
3. Reads the topology from AA.top.
4. Checks that all files are consistent.
5. Creates a binary input file called ions.tpr.
- 6. No ions are added in this step.**
7. It simply prepares the system for the next command.

❑ **Command 2**

1. Reads the prepared system (ions.tpr).
2. Calculates the total charge of the system.
3. Randomly selects water molecules.
4. Replaces those water molecules with Na⁺ and/or Cl⁻ ions.
5. Updates the coordinate file.
6. Updates the topology file with the correct number of ions.



Adding Ions to Neutralize the System

Table 8.1: Breakdown of the first command for neutralizing the system

Command Part	Full Meaning	Function / Purpose
gmx	GROMACS executable	Starts the GROMACS program.
grompp	GROMACS Preprocessor	Combines the simulation parameters, coordinates, and topology into a single binary input file (. tpr) required for the next simulation step.
-f ions.mdp	Parameter File	Specifies the molecular dynamics parameter file (ions . mdp) containing the settings required for ion addition (typically an energy minimization parameter file).
-c AA-solv.gro	Coordinate File	Specifies the input coordinate file containing the solvated protein system.
-p AA.top	Topology	Specifies the system topology file describing the molecules, force field, and system composition.
-o ions.tpr	Output	Creates the portable binary run input file (ions . tpr) used by gmx genion to add ions.



Adding Ions to Neutralize the System

Table 8.2: Breakdown of the second command for neutralizing the system

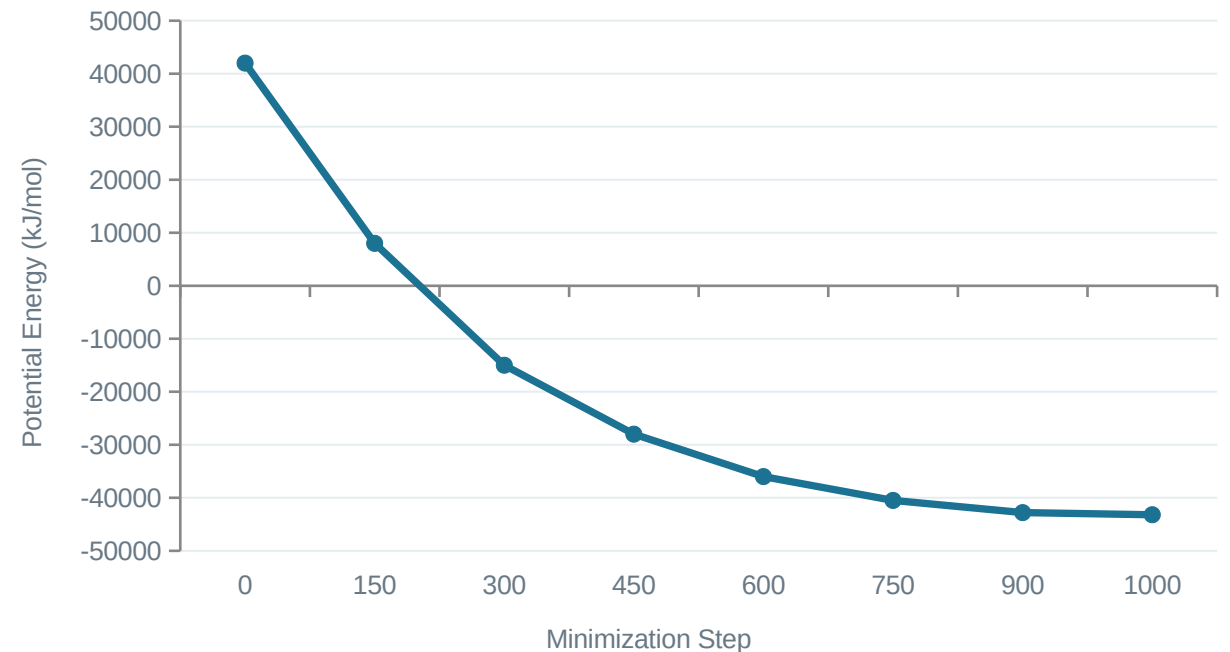
Command Part	Full Meaning	Function / Purpose
gmx	GROMACS executable	Starts the GROMACS program.
genion	Generate Ions	Replaces selected water molecules with ions to neutralize the system or achieve a desired salt concentration.
-s ions.tpr	Simulation Input	Specifies the binary run input file (<code>ions.tpr</code>) created by <code>grompp</code> .
-o AA-solv-ions.gro	Output	Writes the coordinate file containing the protein, water molecules, and added ions.
-p AA.top	Topology	Updates the topology file by modifying the number of water molecules and adding the correct number of ions.
-pname NA	Positive Ion Name	Specifies that sodium ions (Na⁺) will be used as the positive ions.
-nname CL	Negative Ion Name	Specifies that chloride ions (Cl⁻) will be used as the negative ions.
-neutral	Neutralize System	Adds enough Na⁺ or Cl⁻ ions to make the total charge of the system equal to zero.



Energy Minimization

- **Removes** steric clashes and unfavorable contacts introduced while building the system.
- **Steepest descent**, often followed by **conjugate gradient**, are the standard algorithms.
- **Convergence:** run until the maximum force (Fmax) drops below a set threshold.
- **A safety step:** starting equilibration from a high-energy structure can crash the simulation outright.

Illustrative Potential Energy During Minimization





Energy Minimization

❑ Steepest Descent

1. Steepest Descent is the most commonly used energy minimization algorithm.
2. It moves atoms in the direction of the **largest decrease in potential energy** (the steepest downhill path), quickly removing steric clashes and bad contacts.

❑ Conjugate Gradient

1. Conjugate Gradient is often used after Steepest Descent.
 2. It takes more efficient steps toward the minimum energy, making it more accurate for fine optimization once the major clashes have been removed.
- **Potential energy is relative, not absolute.** In molecular dynamics, the potential energy of two atoms infinitely far apart is defined as **0 kJ/mol**. As atoms come closer and form favorable interactions (bonds and attractive forces), the potential energy decreases and can become **negative**, indicating a more stable system.
 - **During energy minimization, atoms move to relieve steric clashes and unfavorable contacts.** As these repulsive interactions are removed, the potential energy decreases from a high positive value to a lower (often negative) value, showing that the structure has reached a more stable, energetically favorable state.



Energy Minimization

❑ **Command 1: `gmx grompp -f minim/em.mdp -c AA-solv-ions.gro -p AA.top -o em.tpr`**

1. Reading the minimization parameters from minim/em.mdp.
2. Reading the coordinates from AA-solv-ions.gro.
3. Reading the topology from AA.top.
4. Checking that all files are consistent.
5. Creating the input file em.tpr for the minimization run.

❑ **Command 2: `gmx mdrun -v -deffnm em`**

1. Reading the input file em.tpr.
2. Applying the minimization algorithm (e.g., Steepest Descent).
3. Moving atoms to remove steric clashes and unfavorable contacts.
4. Reducing the system's potential energy.
5. Writing the minimized structure and simulation output files.



Energy Minimization

Table 9.1: Breakdown of the first command for energy minimization of the system

Command Part	Full Meaning	Function / Purpose
gmx	GROMACS executable	Starts the GROMACS program.
grompp	GROMACS Preprocessor	Combines the simulation parameters, coordinates, and topology into a single binary input file for energy minimization.
-f minim/em.mdp	Parameter File	Specifies the molecular dynamics parameter file (minim/em.mdp) containing the energy minimization settings (e.g., steepest descent algorithm, convergence criteria, maximum number of steps).
-c AA-solv-ions.gro	Coordinate File	Specifies the input coordinate file containing the protein, water molecules, and ions.
-p AA.top	Topology	Specifies the system topology describing the force field, molecules, and interactions.
-o em.tpr	Output	Creates the binary run input file (em.tpr) that contains all information required to perform energy minimization.



Energy Minimization

Table 9.2: Breakdown of the second command for energy minimization of the system

Command Part	Full Meaning	Function / Purpose
gmx	GROMACS executable	Starts the GROMACS program.
mdrun	Molecular Dynamics Run	Executes the simulation or energy minimization using the . tpr input file.
-v	Verbose	Displays detailed progress information, including the potential energy during the run.
-deffnm em	Default File Name	Uses em as the default filename prefix. GROMACS automatically reads em . tpr and generates output files such as em . gro, em . edr, em . log, and em . t r r .



Energy Minimization

- The **.edr (Energy Data)** file stores the energy values calculated during the simulation, such as **potential energy, kinetic energy, total energy, temperature, pressure, and density**. It is used with gmx energy to analyze the energetic and thermodynamic properties of the system.
- The **.trr (Trajectory)** file stores the complete simulation trajectory, including **atomic coordinates, velocities, and forces** at each saved time step. It is a high-precision trajectory file mainly used for restarting simulations and detailed analyses.



Equilibration: Preparing for Production

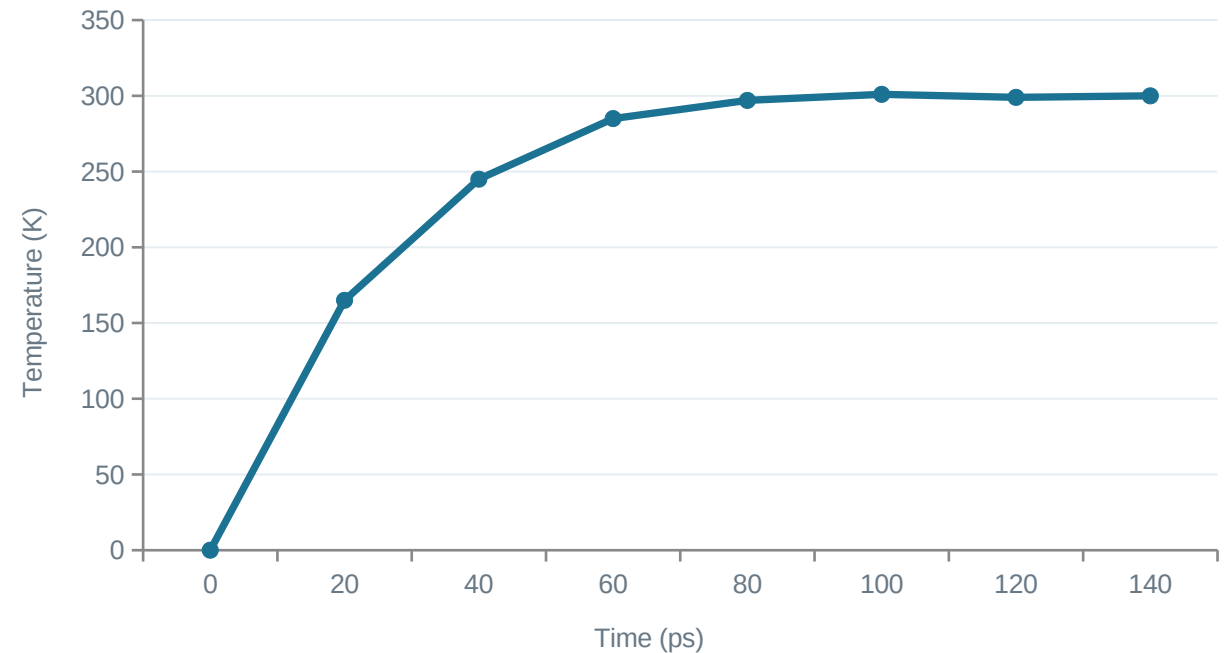
- **Bridges the gap** between a minimized structure and a stable, physically realistic system by gradually stabilizing temperature, pressure, and solvent around the biomolecule.
- **Two stages:** NVT first, then NPT — each holding a different set of quantities constant.
- **Position restraints** are typically applied on heavy atoms (all non-hydrogen atoms) of the protein to maintain its overall structure while allowing surrounding water molecules and ions to move freely, relax, and adapt to the protein.
- **Only once both stages are stable** does the system move on to production.



NVT Equilibration

- **N, V, T held constant** — the number of particles, volume, and temperature.
- **Known as** the canonical ensemble.
- **A thermostat** (e.g., **V-rescale**, **Nosé–Hoover**) drives the system to the target temperature.
- **Typical length:** 100 picoseconds or more, until temperature settles around its target.

Illustrative Temperature Equilibration





NVT Equilibration

Table 10: Difference between V-rescale and Nose-Hoover algorithms

Thermostat	How it Works	Typical Use
V-rescale (Velocity Rescaling)	Maintains the target temperature by gently rescaling atomic velocities while preserving the correct temperature distribution.	Preferred for NVT equilibration because it quickly and stably reaches the desired temperature.
Nosé–Hoover	Maintains the target temperature by coupling the system to an external heat reservoir , allowing natural temperature fluctuations.	Commonly used for production MD and can also be used in NVT (for longer run) once the system is close to equilibrium.

An external heat reservoir is a mathematical heat bath that exchanges energy with the simulation system to maintain the desired temperature while allowing natural temperature fluctuations.



NVT Equilibration

❑ **Command 1: `gmx grompp -f nvt.mdp -c em.gro -p AA.top -o nvt.tpr`**

1. Reading the NVT simulation parameters from `nvt.mdp`.
2. Reading the minimized structure from `em.gro`.
3. Reading the topology from `AA.top`.
4. Checking that all files are consistent.
5. Creating the input file `nvt.tpr` for the NVT simulation.

❑ **Command 2: `gmx mdrun -deffnm nvt`**

1. Reading the input file `nvt.tpr`.
2. Gradually bringing the system to the target temperature using the selected thermostat (e.g., V-rescale or Nosé–Hoover).
3. Keeping the **Number of particles (N)**, **Volume (V)**, and **Temperature (T)** constant.
4. Allowing water molecules and ions to equilibrate around the restrained protein.
5. Writing the equilibrated structure and simulation output files.



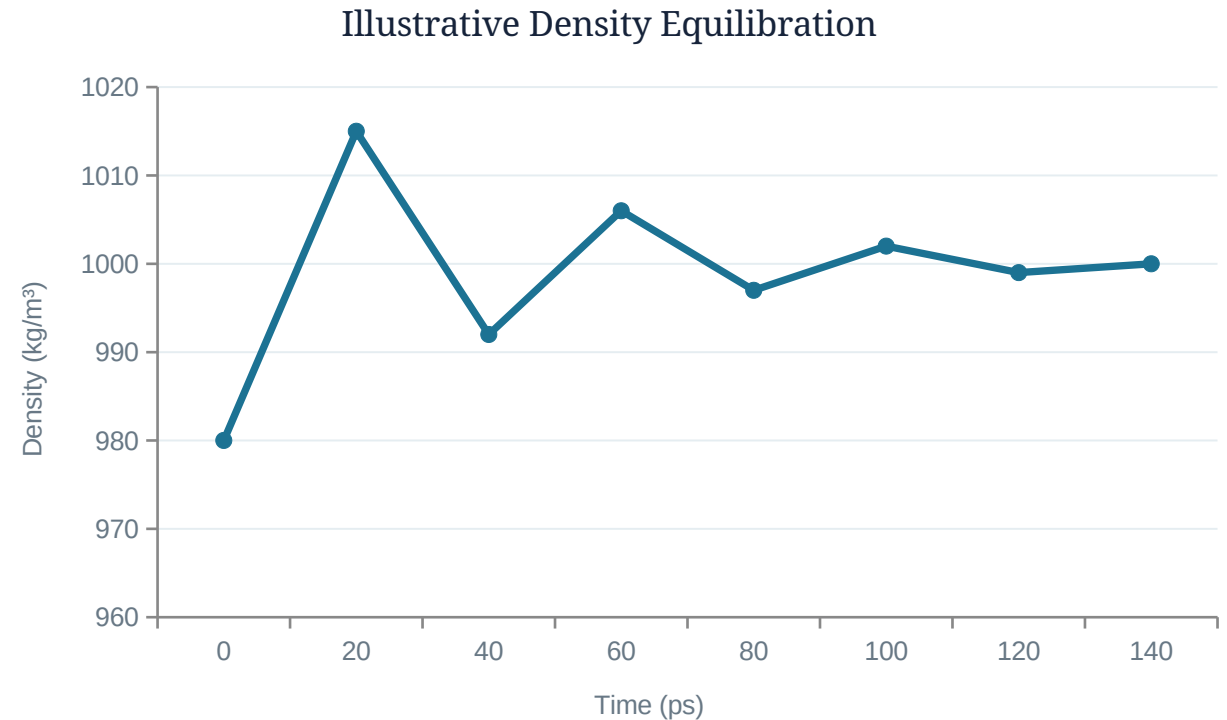
NVT Equilibration

The .cpt (Checkpoint) file stores the current state of the simulation, including coordinates, velocities, simulation time, and other essential information. **If the simulation is interrupted (e.g., due to a power failure or time limit on an HPC cluster),** it allows the simulation to restart from the last saved checkpoint instead of starting over.



NPT Equilibration

- **N, P, T held constant** — the number of particles, pressure, and temperature.
- **Known as** the isothermal–isobaric ensemble.
- **A barostat** (e.g., **Berendsen, Parrinello–Rahman**) drives the system to the target pressure.
- **Confirms** the system has reached the correct, stable density before production begins.





NPT Equilibration

Table 11: Difference between Berendsen and Parrinello–Rahman Barostat

Feature	Berendsen Barostat	Parrinello–Rahman Barostat
How it works	Adjusts the box size smoothly to reach the target pressure.	Allows the box size and shape to fluctuate naturally to maintain the target pressure.
Pressure equilibration	Fast and stable	More realistic but slower
Pressure fluctuations	Suppresses (reduces) natural fluctuations	Produces realistic pressure fluctuations
Accuracy	Good for equilibration	More accurate for production MD
Common use	NPT equilibration	Production MD (and sometimes final NPT equilibration)



Difference between NVT and NPT Equilibration

□ NVT

1. **NVT** stabilizes the system at the **target temperature** while keeping the **volume constant**.
2. During **NPT**, the simulation box is allowed to **expand or shrink** to reach the target pressure.
3. Changes in volume can cause the **temperature to fluctuate**.
4. Therefore, a **thermostat** is used during NPT to maintain the desired temperature while pressure and density equilibrate.

□ NPT

1. In **NVT**, the **volume is fixed**, so the **density remains constant**.
2. In **NPT**, the **volume changes**, causing the density to adjust.
3. The simulation box expands or contracts until the **pressure (reaches to 1 bar) and density become stable**.
4. The final density should be close to the **experimental density** ($\approx 1000 \text{ kg/m}^3$ for water at room temperature).



NVT and NPT Equilibration

❑ NVT

Command 1: `gmx grompp -f nvt.mdp -c em.gro -p AA.top -o nvt.tpr`

Command 2: `gmx mdrun -v -deffnm nvt`

❑ NPT

Command 1: `gmx grompp -f npt.mdp -c nvt.gro -t nvt.cpt -p AA.top -o npt.tpr`

Command 2: `gmx mdrun -v -deffnm npt`



Production MD

- **Position restraints** are fully released — the system now evolves freely.
 - **This is the run** that actually generates the trajectory used for analysis.
 - **Length depends on the question:** nanoseconds for local motion, microseconds or beyond for larger conformational change.
 - **Typical analyses:** RMSD, RMSF, radius of gyration, hydrogen-bond occupancy, and interaction energies.
- ❑ **Command 1:** `gmx grompp -f md.mdp -c npt.gro -t npt.cpt -p AA.top -o md.tpr`
 - ❑ **Command 2:** `gmx mdrun -v -deffnm md`



Analysis-RMSD (Root Mean Square Deviation)

- **Root Mean Square Deviation (RMSD)** measures the **average structural deviation** of a protein from a reference structure (usually the initial structure) during a molecular dynamics simulation.
- Evaluates the **structural stability** of a protein over time.
- Monitors **conformational changes** during the simulation.
- Determines whether the system has reached **equilibrium**.

RMSD Behavior	Interpretation
Stable plateau	Protein has equilibrated and remains structurally stable.
Gradual increase, then plateau	Protein undergoes conformational adjustment before stabilizing.
Continuous increase	Protein may still be changing conformation or has not reached equilibrium.
Large fluctuations	Indicates high structural flexibility or possible instability.

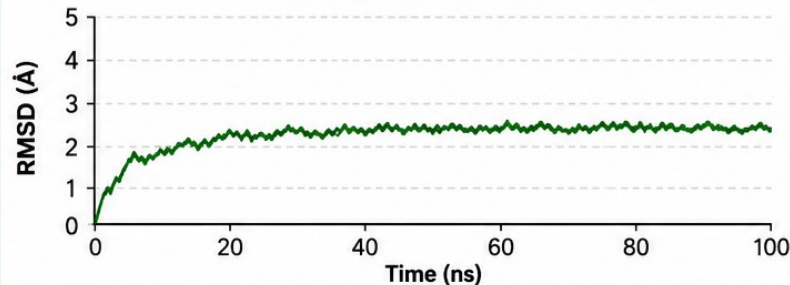


Analysis-RMSD (Root Mean Square Deviation)

RMSD Behavior and Interpretation

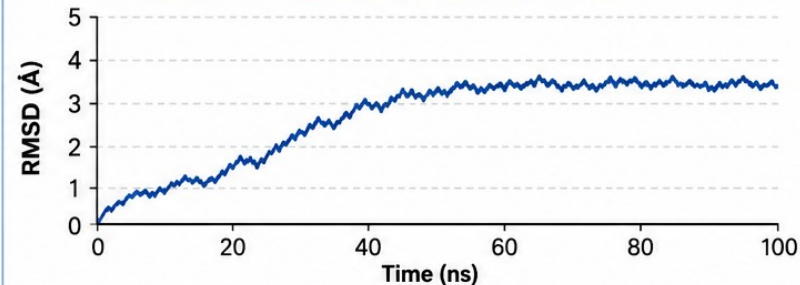
RMSD (Root Mean Square Deviation) measures the average deviation of a protein structure from a reference structure during MD simulation.

1. Stable Plateau



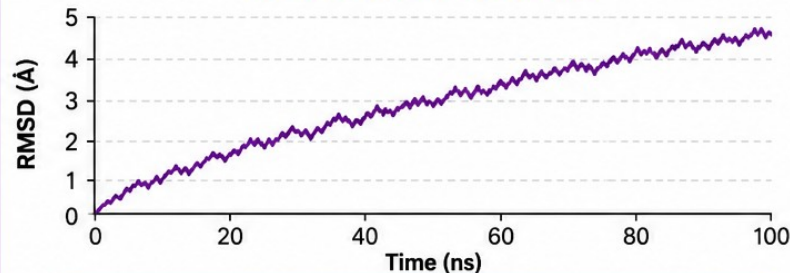
Interpretation: Protein has equilibrated and remains structurally stable.

2. Gradual Increase, Then Plateau



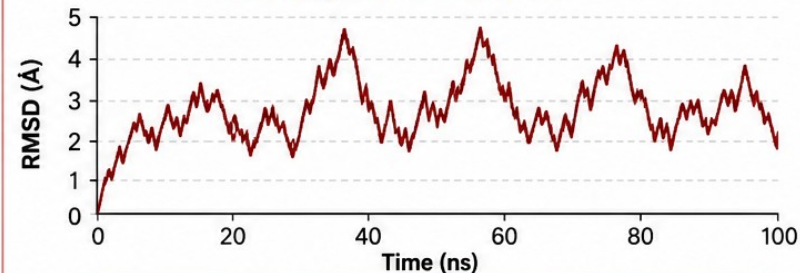
Interpretation: Protein undergoes conformational adjustment before stabilizing.

3. Continuous Increase



Interpretation: Protein may still be changing conformation or has not reached equilibrium.

4. Large Fluctuations



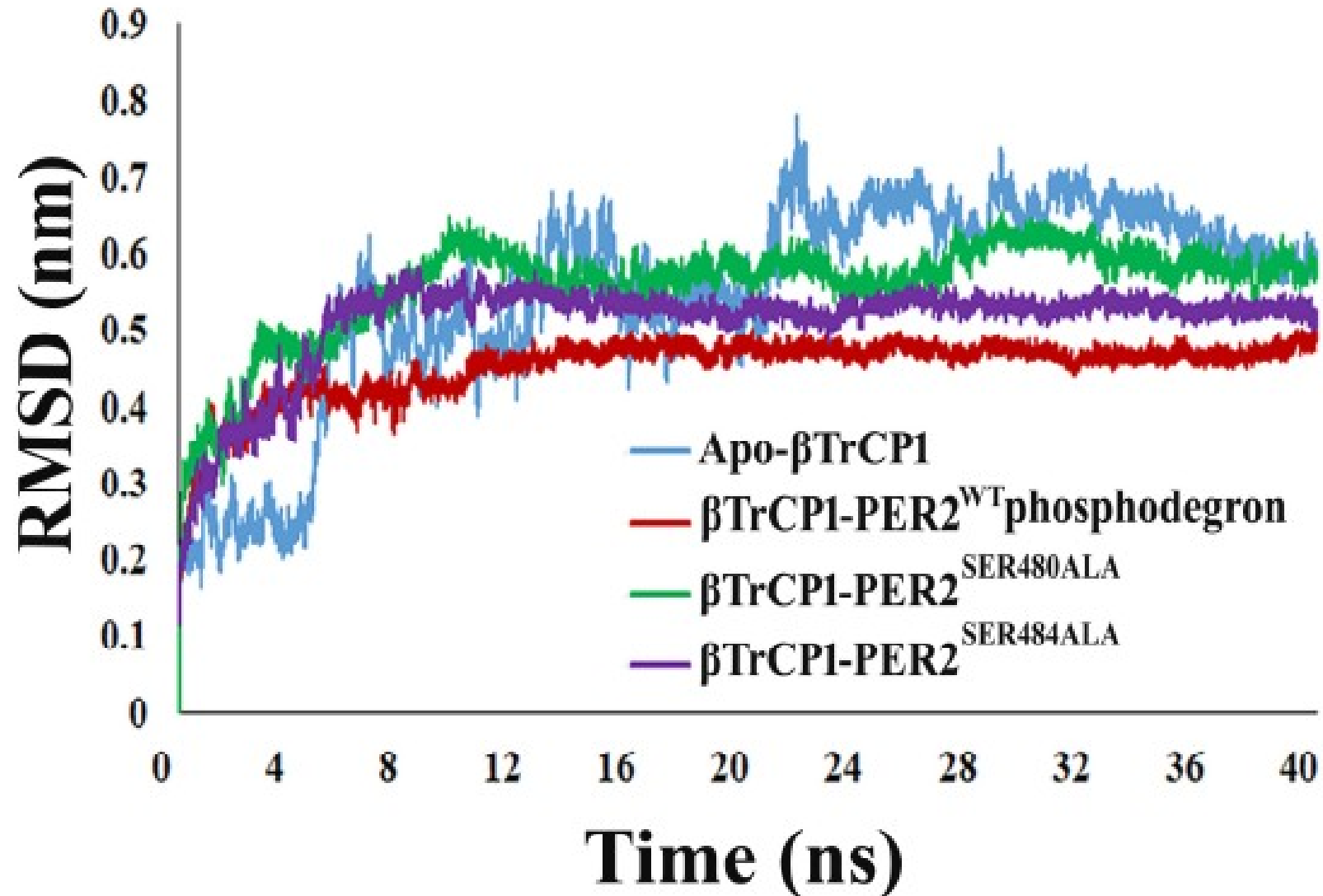
Interpretation: Indicates high structural flexibility or possible instability.



Take-Home Message: RMSD helps evaluate the structural stability of a protein during MD simulation. A stable plateau suggests equilibrium, while increasing or highly fluctuating RMSD indicates ongoing changes or instability.



Analysis-RMSD (Root Mean Square Deviation)





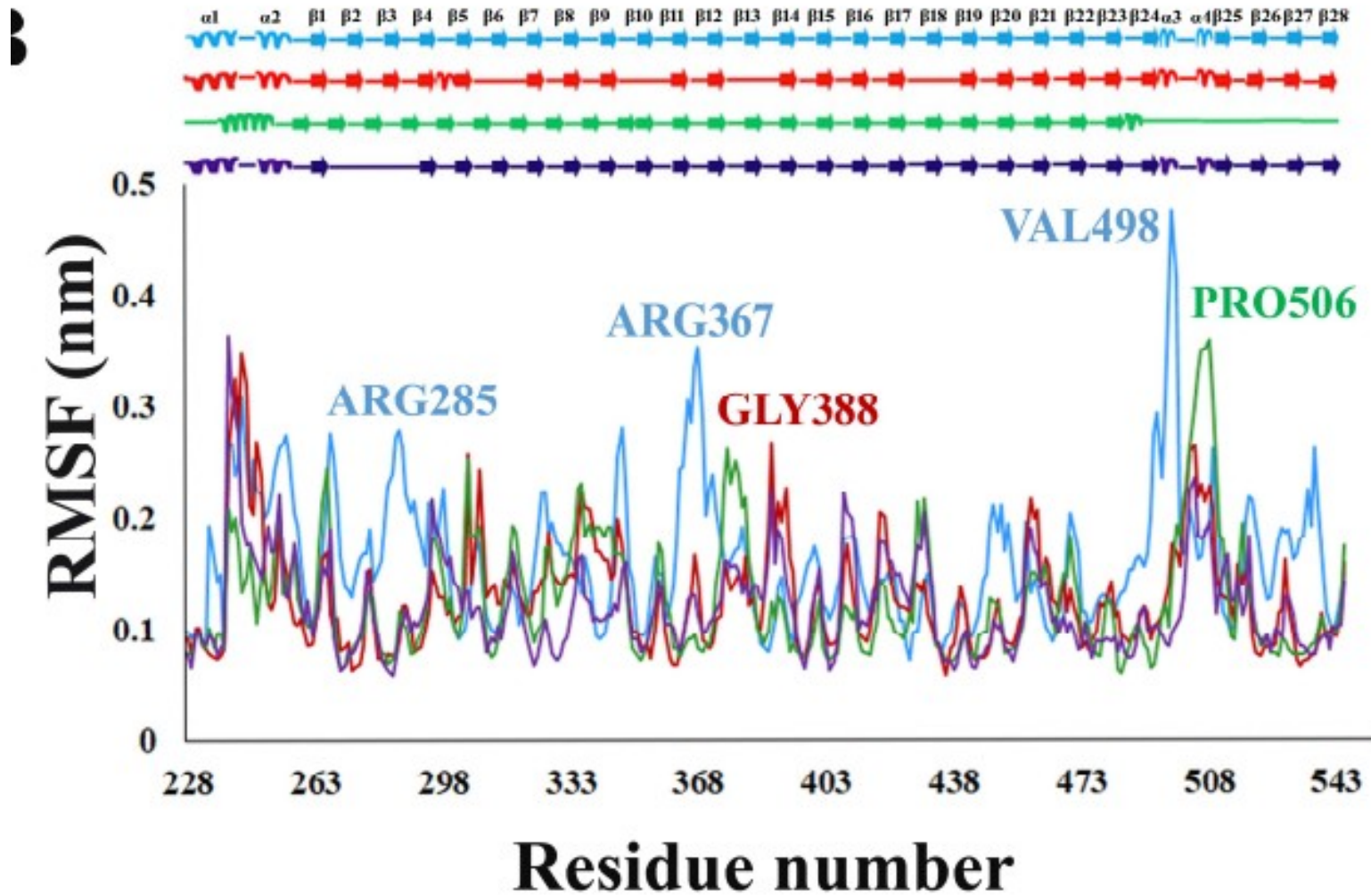
Analysis-RMSF (Root Mean Square Fluctuations)

- **Root Mean Square Fluctuation (RMSF)** measures the **average fluctuation of each atom or residue around its mean position** during a molecular dynamics simulation. It identifies the **flexibility of different regions** of a protein.
- Identifies **flexible and rigid regions** of a protein.
- Detects residues involved in **binding, loops, and domain movements**.
- Compares the flexibility of **wild-type and mutant proteins** or **protein–ligand**

RMSF Value	Interpretation
Low RMSF	Residues are stable and rigid (often α -helices and β -sheets).
Moderate RMSF	Indicates normal flexibility required for protein function.
High RMSF	Residues are highly flexible (commonly loops and terminal regions).
Localized high peaks	May indicate conformational changes or binding-induced flexibility.

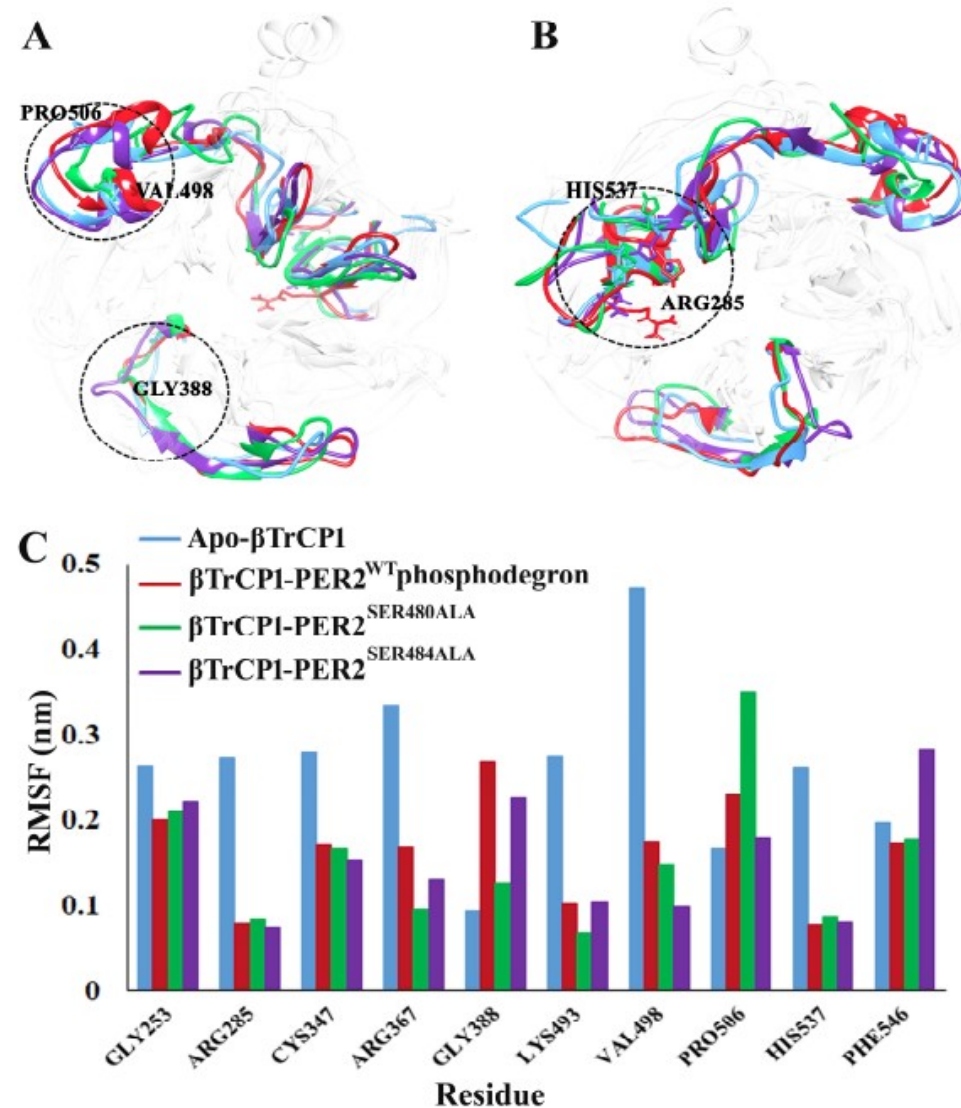


Analysis-RMSF (Root Mean Square Fluctuations)





Analysis-RMSF (Root Mean Square Fluctuations)





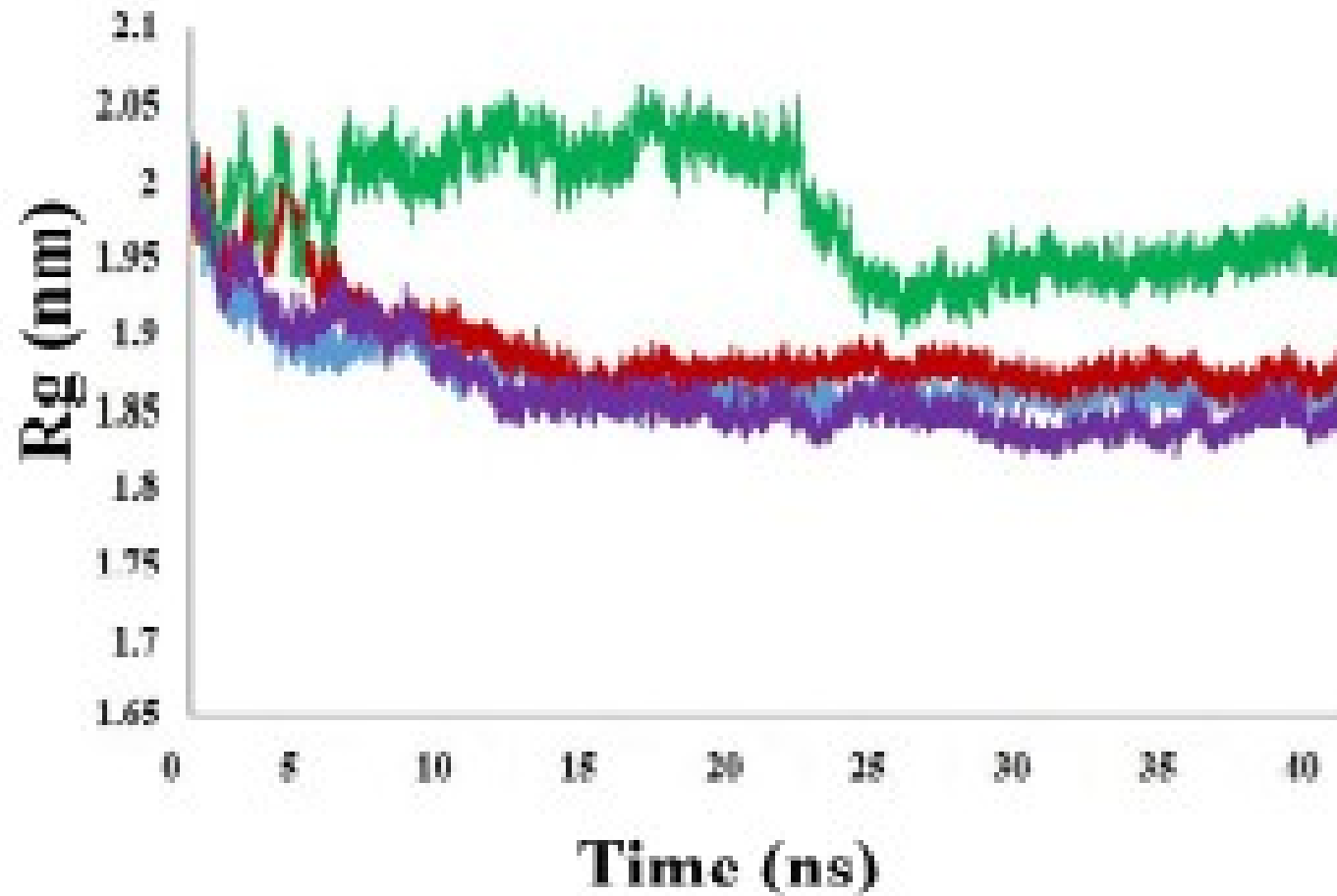
Analysis-Rg (Radius of Gyration)

- **Radius of Gyration (Rg)** measures the **overall compactness of a protein** by calculating how its atoms are distributed around the protein's center of mass during a molecular dynamics simulation.
- Evaluates the **compactness and structural stability** of a protein.
- Detects **folding, unfolding, or expansion** during the simulation.
- Compares the compactness of **wild-type vs. mutant** proteins or **apo vs. ligand-bound**

Rg Behavior	Interpretation
Stable plateau	Protein remains compact and structurally stable.
Gradual decrease	Protein becomes more compact (folding or tighter packing).
Gradual increase	Protein becomes less compact (expansion or partial unfolding).
Large fluctuations	Indicates structural instability or major conformational changes.



Analysis-Rg (Radius of Gyration)





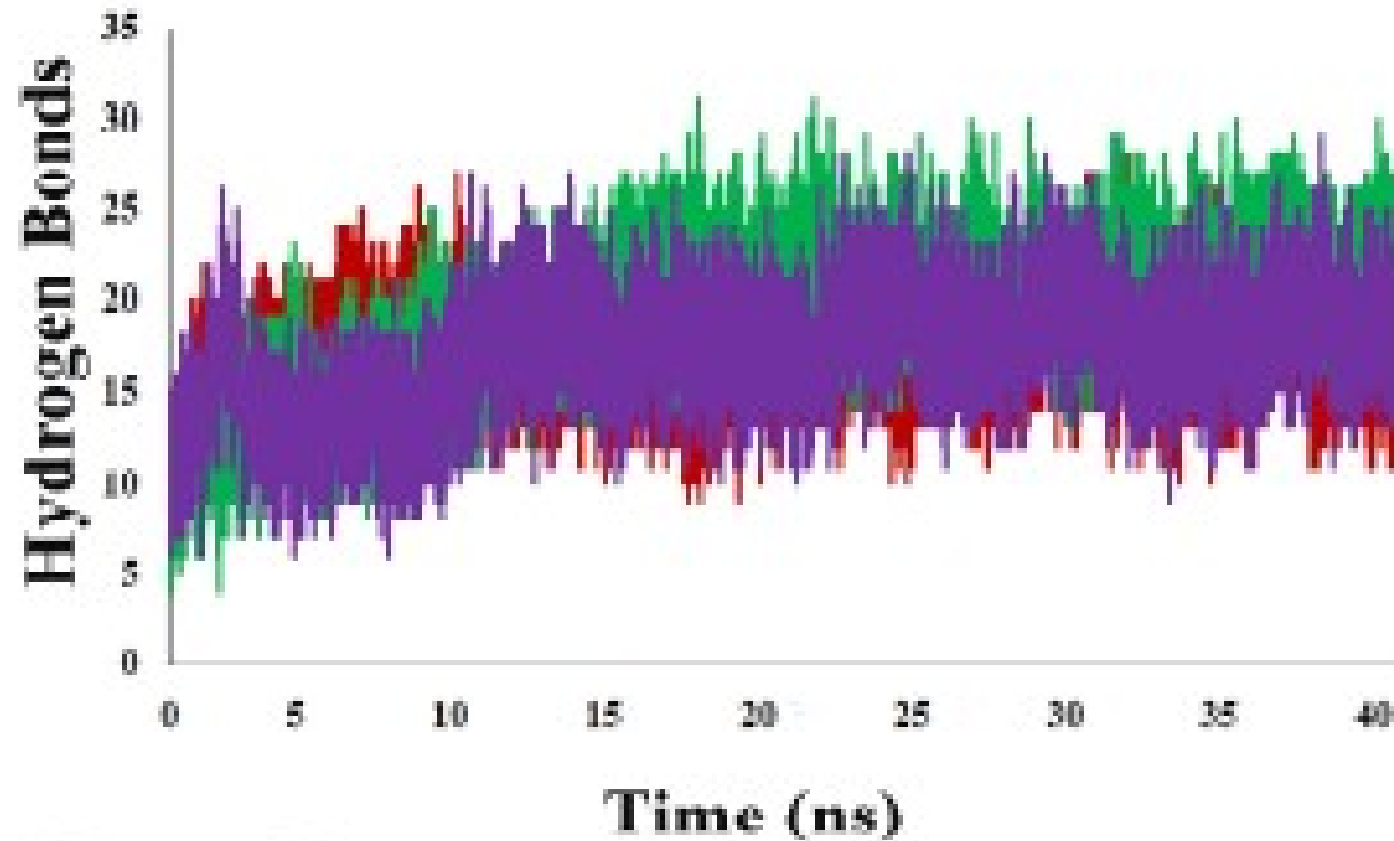
Analysis-Hydrogen Bond

- A **hydrogen bond** is a **weak, non-covalent interaction** formed when a hydrogen atom covalently bonded to an electronegative atom (such as **O** or **N**) is attracted to another electronegative atom nearby.
- Stabilize the **3D structure** of proteins.
- Maintain **protein–ligand** and **protein–protein** interactions.
- Contribute to the stability of **DNA base pairing**.
- Influence **binding affinity** and **molecular recognition**.

Hydrogen Bond Behavior	Interpretation
High and stable number	Strong and stable molecular interactions.
Moderate fluctuations	Normal dynamic formation and breaking of hydrogen bonds.
Continuous decrease	Loss of structural stability or weaker binding.
Large fluctuations	Flexible or unstable interactions.



Analysis-Hydrogen Bond



SECTION 2 OF 3

Molecular Dynamics Simulation of Protein–Drug Interactions

Extending the same pipeline to systems where a small-molecule drug is bound to the protein.





What Changes with a Bound Ligand?

- **Standard force fields** (AMBER, CHARMM, GROMOS, OPLS) don't include parameters for drug-like small molecules.
- **The ligand needs its own topology:** atom types, partial charges, and bonded parameters, generated separately.
- **Protein and ligand topologies** are then merged into a single, consistent system.
- **Everything after that** — box, solvation, ions, minimization, equilibration, production — follows the same pipeline as Section 1.



Ligand Parametrization Tools

Protein force fields don't cover drug-like small molecules — a short parametrization step fills that gap.
Whichever tool is used, the ligand and protein force field must match

The two topologies are then merged into a single, consistent system before solvation.

CGenFF

CHARMM-compatible ligand parameters,
via the CGenFF web server

ACPYPE/ Antechamber

AMBER-compatible parameters, built on
GAFF

LigParGen

OPLS-AA-compatible ligand parameters

ATB (Automated Topology Builder)

GROMACS compatible parameters

SECTION 3 OF 3

CHARMM-GUI: An Easy Path to System Preparation for MD Simulation

Building the same simulation-ready system through a browser instead of the command line.





Introduction to CHARMM-GUI

- **CHARMM-GUI** is a free, web-based interface for building simulation-ready systems (charmm-gui.org).
- **Removes the need** to type each preparation command by hand.
- **Generates input** not only for GROMACS, but also NAMD, AMBER, CHARMM, and OpenMM.
- **Purpose-built modules** for many system types — Solution Builder, Membrane Builder, Glycan Reader, Ligand Reader & Modeler, and more.
- **A popular starting point** for newcomers, and a real time-saver for complex systems.



Reading Biomolecular Coordinates

- **Upload a PDB file directly**, or fetch a structure straight from the RCSB PDB by its ID.
- **Automatically detects** chains, hetero groups, and missing residues or loops.
- **Lets you choose** which chains, subunits, or bound ligands to carry forward.
- **Assigns** — or lets you adjust — residue protonation states before building continues.

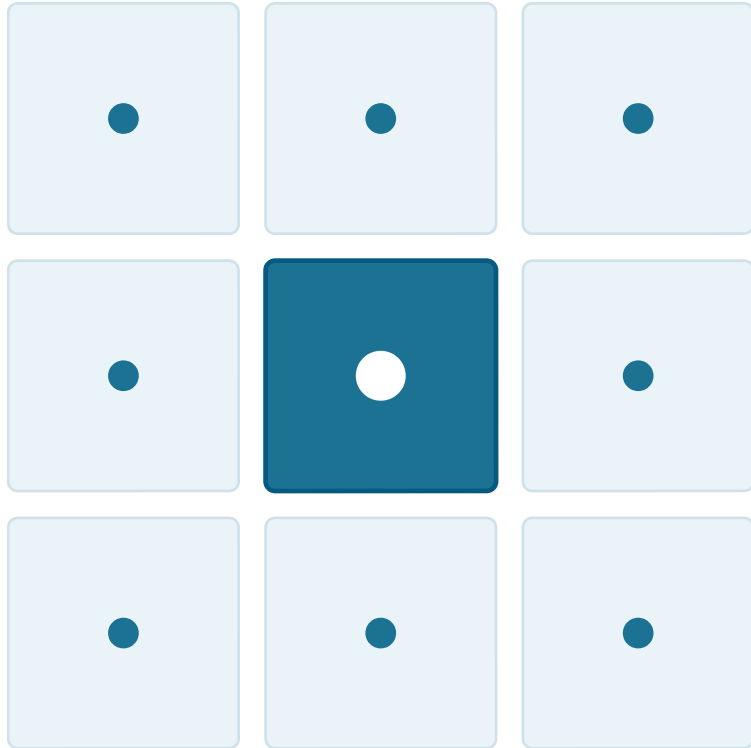


Solvating the Biomolecule

- **Water model** — TIP3P, TIP4P, or SPC — chosen from a simple menu.
- **Box shape and edge distance** set the same way, without typing a single flag.
- **Ion type and concentration** are added directly — no separate genion step required.
- **Every choice** from Section 1's command-line workflow is available here as a form field.



Setting Periodic Boundary Conditions



-  Simulated box — what GROMACS actually computes
-  Periodic images — mathematical copies, not real atoms

- **A finite box** can't represent a bulk solvent environment on its own.
- **PBC treats the box** as if it were tiled infinitely in every direction.
- **An atom leaving one side** reappears at the opposite side, and can interact across that boundary.
- **Built in automatically** by both editconf/solvate on the command line and by CHARMM-GUI.



Equilibration and Production

- **Ready-to-run scripts** and .mdp files are written out for every stage automatically.
- **A multi-step equilibration** is pre-configured, with restraints released gradually.
- **Production-run scripts** are generated in the very same download.
- **Run the numbered steps in order:** minimization, then equilibration, then production.

Two Paths, Same Destination



Command Line

- pdb2gmx, editconf, solvate, and genion, typed one at a time
- Full control over every flag and parameter
- The standard route in most published GROMACS protocols
- A steeper learning curve for first-time users



CHARMM-GUI

- The same stages, built through a guided web form
- Faster to a working system, especially for complex builds
- Strong defaults for membranes, glycans, and other advanced systems
- Still worth understanding the command-line steps underneath

Both paths converge on the same production-ready GROMACS system.



Key Takeaways

- **One pipeline:** topology, box, solvate, neutralize, minimize, equilibrate, produce.
- **One extra step for drugs:** parametrize the ligand, then merge its topology with the protein's.
- **Two ways to get there:** the GROMACS command line and CHARMM-GUI build the same kind of system.
- **Always check convergence:** confirm temperature, pressure, and density have stabilized before trusting production data.

The pipeline, once more:



Thank You

Questions & Discussion

[Presenter email / contact]

